

Smart OCR for Recognizing Bangla Characters with CRAFT and Deep Learning Models

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Abstract – Optical Character Recognition (OCR) is a popular technology for a variety of applications. The purpose of implementing optical character recognition is to turn printed text into editable text. OCR accuracy can be influenced by text preparation and segmentation techniques. Occasionally, it might be difficult to obtain text from an image due to its varying size, style, orientation, and intricate backdrop, among other factors. If the printed or handwritten texts are in Bangla font, the work gets more difficult. Because Bangla language has 50 characters and 24 scrape characters (প্রতিবর্ণ) for example, অ-কার/আ-কার (া), ই-কার/ঈ-কার (ি/ী), উ-কার/ঊ-কার (ু/ূ) খ-কার (্), ব-ফলা (্‌ব), ম-ফলা (্‌ম), য-ফলা (্‌য), র-ফলা (্‌র) etc. Moreover, in bangla language there are several diphthong or compound letter (যুক্তাক্ষর) for example, জ, ঝ, ঞ, ঠ, ঙ etc. Therefore, even though there are several OCR-related works really good for the English font, for Bangla, we do not have without NLP (Natural Language Processing). So here We have tried to cover both the English and the Bangla font with our proposed model using crafting, VGGNet and various other machine learning algorithms such as LSTM and CTC.

Index Terms— *Optical Character Recognition (OCR), crafting, Bangla font OCR, Handwritten Text Support, Bangla Scrape Characters, Compound Character, LSTM, CTC, Pytorch, VGGNet, Deep Learning (DL) model.*

I. INTRODUCTION

Optical character recognition (OCR) technology turns text input into a machine-encoded format. Today, OCR aids not just in the digitization of medieval handwritten manuscripts, but also in the conversion of typewritten materials to digital format. This has made retrieving essential information easier since one no longer needs to go through mounds of paperwork and files to get the information. The demands for the digital preservation of historical data, legal records, instructional materials, etc., are being met by organizations. An OCR system is primarily dependent on the extraction of characteristics and their discrimination/classification (based on patterns). As a branch of OCR, handwritten OCR has a growing amount of attention. In the 1940s, one of the first OCR systems was established; as

technology improved, the system grew increasingly capable of handling both printed and handwritten characters, and OCR devices eventually became commercially available. However, there is significant interest in optical character recognition systems that can read handwritten Bangla fonts. Our national government has begun the process of digitizing records from all of the most crucial sectors. But if we know where to look, we may find a wealth of information and records that have been meticulously recorded by hand. Since digitalization is already occurring, it is no longer sufficient to keep simply the records of the current generation digitalized; all the material from the past also needs to be updated in a digital form. It is no secret that Bangla fonts are not as user-friendly as their English counterparts. There may be an overwhelming number of compound letters in a Bangla font [1].

Traditional current NLP-based OCRs for English typefaces have difficulty deciphering these complex letters. For this reason, we will now try the concept of crafting with CNN and other machine-learning algorithms. “Character-Region Awareness For Text detection”, or CRAFT, is an efficient method of locating text areas by analyzing the affinity between characters and the regions of those characters. After thresholding the character region and affinity scores, we can easily locate the minimal bounding rectangles on the binary map to obtain the text's bounding box. And the recent breakthroughs of deep learning systems in image recognition may be attributed in great part to automatic feature selection. Features from a scale-invariant feature transform and accelerated robust features are examples of the kinds of features used by conventional image recognition algorithms. In order to carry out image recognition, the gathered characteristics are fed into machine learning algorithms. It follows that the performance of a standard image recognition system is heavily dependent on the attributes collected from a picture. However, feature extraction is laborious and time-consuming. When trying to use a solution for more than one problem or purpose, it is usually necessary to redesign some of the features. This procedure produces artificial characteristics. Automatic feature extraction using CNN may now be done without the requirement for any form of feature design. CNN feature

extracted approaches have conquered over more traditionally constructed feature extraction algorithms in a number of ImageNet image recognition competitions. Therefore, we think CNN is the best way to identify the handwritten Bengali fonts.

II. BACKGROUND

Optical character recognition (OCR) is a system that is dependent on the extraction of features which are then further used for classification. In optical character recognition (OCR), the handwritten area of it has generated a lot of curiosity. The data received can be broken down into two systems that is offline system and online system. For the offline process we have to scan the desired documents for it to process the document and generate output and the online process depends on a lot of things, it depends on how we write the text, the position of the text, in which angle the text are written, etc. Thus, the online processing of text is far more complicated than offline process. We have taken only offline input taking method for our research and modified pre-trained CRAFT model which is for English Characters only and was developed by the Cornell University [2]. Our deep learning execution is based on Pytorch. We worked on python 3.9 version though the pre-trained CRAFT model was on 2.7 kernel which doesn't support torch. So that we had to re-model it according to our version.

The recognition model that we have considered here is a Convolutional Recurrent Neural Network (CRNN) which has 3 main segments.

- Feature extraction (VGGNet)
- Sequence labeling (LSTM)
- Decoding (CTC).

We have taken into consideration of a modified version of the training pipeline for performing recognition which is the deep-text-recognition-benchmark framework [3]. Beam search code idea and resources are considered from this [4]. Data synthesis is based on TextRecognitionDataGenerator [5]. For the text analysis using OCR, firstly, the image data we feed is transformed to binary data. The scanned data is then divided into a black and white picture where the white parts are the background and the black parts are the text. Then it moves onto data pre-processing where the scanned data is cleaned and any noise is removed or straightening and smoothing texts in the scanned jpeg/png image. Then the pre-processed data we get is detected. Pattern matching and feature extraction is done on the cleaned image data. During the pattern matching every single word is matched from the image with the stored data (dictionary) we have and in feature extraction phase using the extracted image data we identified or classified the words using VGGNet and machine learning algorithms from the training dataset. Finally the image document is decoded and post-processed where the data is further cleaned and output is generated.

III. METHODOLOGY

A. Proposed Methodology

In this paper, we propose a model that uses a pre-trained model of CRAFT from Cornell University [2] which is trained for English character only. We have modified its bounding box (by multiplying 0.80) to capture compound words of Bangla words as ground truths. The pipeline is like the below diagram.

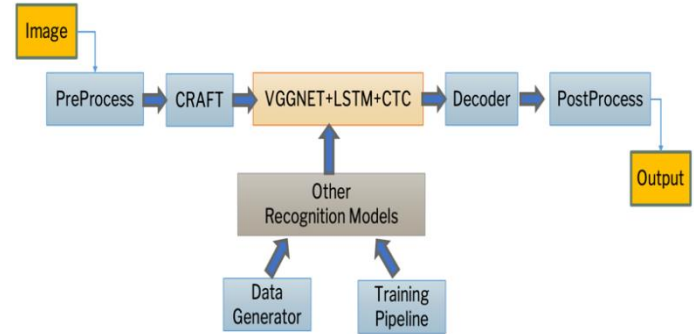


Figure 1. Architecture model to recognise characters

Here we give an input image as our file extension supports png, jpg, tiff and the image file size limit is 2 Mb maximum. And the image dimension limit is 1500 pixel. For recognizing a character ['ch_sim','bn'] that executes the languages we determine to read. We can pass several languages at once by using ['ch_sim','bn', 'en', 'x', 'y'] where 'bn' is for Bangla, 'en' is for English, and 'x' and 'y' are for x, y languages respectively.

```

import easyocr
reader = easyocr.Reader(['bn_sim','bn'])
result = reader.readtext('bangla.jpg')
  
```

But in our case we pass Bangla and English together only as we have only these two character list and dictionary defined. Instead of the filepath bangla.jpg, As an OpenCV, we can also pass an image object. The line we have used here, `reader = easyocr.Reader(['bn_sim','bn'])` is for loading a model into memory. It takes some time but it needs to be run only once. We set detail=0 for simpler output. And we need GPU support to run our training.

```
reader.readtext('bangla.jpg', detail = 0)
```

```
$ easyocr -l bn_sim en -f bangla.jpg --detail=1 --gpu=True
```

VGGNet will work as feature extractor, for sequence labelling we have used LSTM and CTC for the decoding part.

B. Pre-Trained Model (CRAFT)

We have implemented a FC convolutional network (fully connected) based on VGG-16 [6] model as our backbone.

Our model has skip connections in the decoding part, which is similar to U-net [7] in that it adds up low-level features. The final output has two channels as score maps,

1. Region score
2. Affinity score

The network architecture is illustrated in Figure 2. We have adopted batch normalization technique in VGGNet.

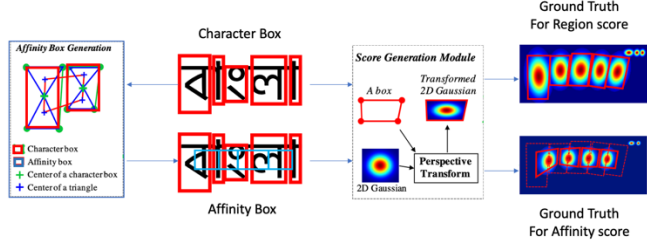


Figure 2. Setting Ground Truths (GTs) based on region and affinity scores

For each training image, we build character-level bounding boxes for the ground truth label for the region score and affinity score. The region score indicates the likelihood that a specific pixel is the character's center, whereas the affinity score indicates the likelihood that a specific pixel is the center of the area between two neighboring characters.

B. Proposed Dataset

In our research, we propose a new customized dataset, which has 2 basic parts.

- Character List (.char)
- Word Dictionary (.dict)

In our dataset, we have 2 languages support; English and Bangla. In table 1 the dataset details are shown.

TABLE 1 PROPOSED DATASET		
	English	Bangla
CHARACTERS	52	74
DICTIONARY WORDS	57,767	60,681

For English characters (en.char) we have considered, $\alpha_{capital}$ and α_{small} for capital and small

letters respectively. We have given an txt file (en.dict) for training and defining ground truths for the words.

$$\alpha_{capital} == \{(A, B, C, D \dots \dots Z)\} \quad 1$$

$$\alpha_{small} == \{(a, b, c, d \dots \dots z)\} \quad 2$$

For English characters (bn.char) we have considered, 50 characters {অ, আ, ক, খ...ইত্যাদি} and we have also

inserted 24 more characters in the (bn.char) list for the scrape characters(প্রতিবর্ণ) { অ-কার/আ-কার (া), ই-কার/ঈ-কার (ি/ী), উ-কার/ঊ-কার (ু/ূ), ঋ-কার (ৃ), ব-ফলা (্‌ব), ম-ফলা (্‌ম), য-ফলা (্‌য), র-ফলা (্‌র) ইত্যাদি } We have given an txt file (bn.dict) for training and defining ground truths for the words.

42	ঈ	শীতকালেই	13	M	86	hungry
43	ঈ	শঙ্কিল	14	N	87	including
44	ঘ	জাগিয়াছিলে	15	O	88	time
45	ঈ	চুড়ি-দুল	16	P	89	being
46	ষ	ঐনিবাস	17	Q	90	there
47	ঊ	জিয়াব	18	R	91	before
48	ফ	জলযজ্ঞ	19	S	92	years
49	ছ	জানেশ্বর	20	T	93	American
50	ল	চমবে	21	U	94	later
51	জ	শিল্পকলার	22	V	95	both
52	আ	সিটি-লিভারপুলের	23	W	96	until
53	।	স্বৈরাচারী	24	X	97	him
54	উ	লিখবে	25	Y	98	against
55	ই	বুলডগ	26	Z	99	while
56	স	রাজস্থান	27	a	100	After
57	ঐ	ঘটিল	28	b	101	released
58	ঊ	ঘটনাবলি	29	c	102	University
59	ঋ	লভলে	30	d	103	She
60	ৱা	উঁচিয়ে	31	e	104	school
61	ূ	ছাড়া	32	f	105	played
62	ৃ	হৈ	33	g	106	As
63	ঌ	লুটিয়াছিল	34	h	107	second
64	৆	পর্জা	35	i	108	called
65	ে	কোড়িয়াছিলি			109	well
		চোকা			110	no
		কুটুহিতছিলে			111	number
		শোকাই			112	will
		সাহিত্যাকাশ			113	early
		ইকাইডেন			114	film
		গ্রেস্‌টার?			115	team
					116	several

Figure 3. Dataset sample with character list and dictionary

IV. RESULT AND DISCUSSION

A. Confidence Score

For Confidence score evaluation, we have considered 50 characters of bangla language, {অ, আ, ই, ঈ, উ, ঊ, ঋ, ঌ, ঍, ও, ঔ} and {ক, খ, গ, ঘ, ঙ, চ, ছ, জ, ঝ, ঞ, ট, ঠ, ড, ঢ, ণ, ত, থ, দ, ধ, ন, প, ফ, ব, ভ, ম, য, র, ল, শ, ষ, স, হ, ড়, ঢ়, ঝ, ঞ, ঔ, ঐ, ঊ} We can calculate confidence score with our equation 3.

Let us assume that we have a sample b of the training data and that K(b) and l(b) represent the bounding box's region and the sample b's length, respectively. By dividing the word's characters, the crafting portion of the process now begins. By using the formula, we can determine the estimated character bounding boxes and their respective character lengths $l^a(b)$. Then our approximate confidence score is denoted as $C_{conf}(b)$ for the sample b is computed as in equation 3,

$$C_{conf}(b) = \frac{l(b) - \min(l(b), |l(b) - l^a(b)|)}{l(b)} \quad 3$$

and the confidence score map with respect to pixel C_{conf} , for an image is computed as in equation 4,

$$C_{conf'}(pixel) = \begin{cases} C_{conf}(b) & \text{pixel} \in K(b) \\ 1 & \text{otherwise} \end{cases} \quad 4$$

B. Testing with Images

At first we have given input the compound Bangla words to test our model Figure 4. Though it was not able to capture the lower letter in compound word from the beginning, then we have increased the bounding box parameter to capture the compound word.

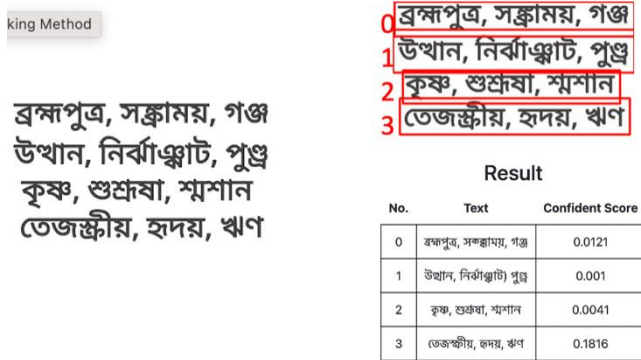


Figure 4. Capturing the compound words with confident scores

Then for the 2nd type testing, we have combined English and Bangla words mixed in a sentence with combined words of Bangla compound letters. And our model successfully can recognize the letters with more than 90% confident scores. Point to be noted, in Figure 5 unfortunately the word “CSE” is detected as “SE”. So we need to look into these matter in our future work.

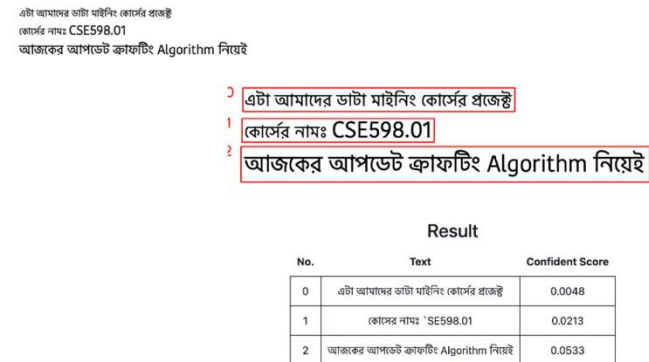


Figure 5. Recognizing Bangla-English mixed sentence words.

Now for the 3rd type testing, we have used real life street sign photos as inputs. Our model can recognise those texts from the image with notably low confident scores due to the noise present on those photos. For example, in Figure 6 ‘চলাচল’ detected as ‘চলচিল’, ‘মোড়’ detected as ‘মৌড়’ etc.

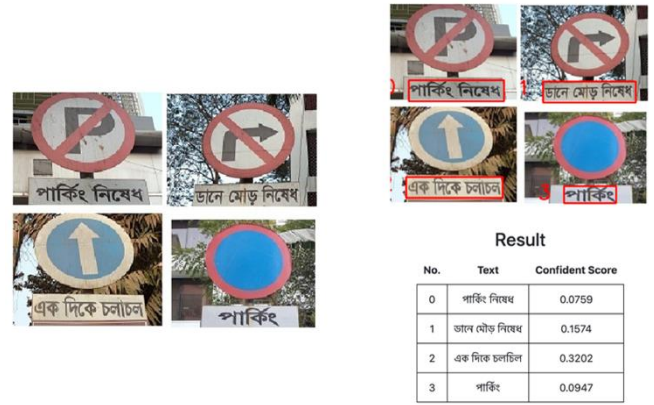


Figure 6. Testing with real life street captured image

Now its time for testing the 4th type, handwritten photos. One of our group-mate gave this handwritten image as input and here is our output with 14 bounding boxes (0-13) and confident score in Figure 7.

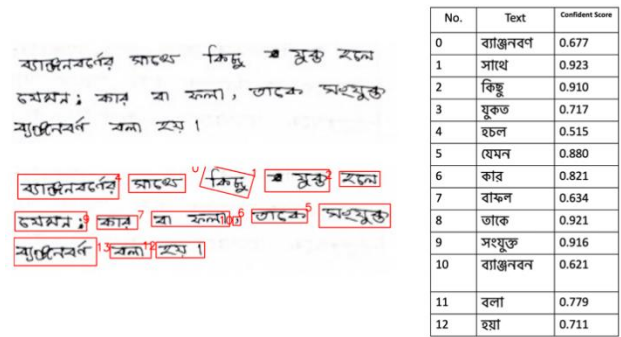


Figure 7. Capturing the compound words with confident scores

V. CONCLUSION

In this research, we have developed an OCR model and we have tried the concept of crafting with CNN (VGGNet) and other machine learning algorithms (LSTM, CTC). “Character-Region Awareness For Text detection”, or CRAFT, which we have implemented from a paper of Cornell University [2] is an efficient method of locating text areas by analyzing the affinity between characters and the regions of those characters. After thresholding the character region and affinity scores, we are able to locate the minimal bounding rectangles on the binary map to obtain the text's bounding box. And the recent breakthroughs of deep learning systems in image recognition may be attributed in great part to automatic feature selection. Features from a scale-invariant feature transform and accelerated robust features are examples of the kinds of features used by conventional image recognition algorithms. In order to carry out image recognition, the gathered characteristics are fed into machine learning algorithms. It follows that the performance of a standard image recognition system is heavily dependent on the attributes collected from a picture. Our future plan is to try ResNet50 for better feature detection and

knowledge distillation to consume less time in training as teacher model will be compressed into student model to run the model with less resources and small model size.

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